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PROBABILITY DISTRIBUTIONS

STEAM-3D

Sustainable Training in Epidemiological Analysis and Modelling
for Data-Driven Decision Making in Africa

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Sustainable Training in Epidemiological Analysis and Modelling for Data-Driven Decision Making in Africa

SOUTH AFRICA | MALAWI | MOZAMBIQUE | CAMEROON | BELGIUM

Learning Outcomes

- Understand probability distributions
  - Distinguish discrete and continuous
  - Application
- Binomial distribution
- Poisson distribution
- Normal (Gaussian) distribution
- Standard Normal distribution

Probability Distributions

- Formula or a table used to assign probabilities to each possible value of a random variable X.

- Discrete
  - X can assume one of a countable (usually finite) number of values
  - e.g. binomial, Poisson, hypergeometric, etc.

- Continuous
  - X can assume one of an infinite (uncountable) number of different values
  - e.g. normal (Gaussian), logistic, Gamma

Binomial Distribution

- If an event can have only two possible outcomes (yes/no, positive/negative), the outcome is considered *binary* with either A or B as possible outcomes.

- Probability of A is denoted by π. Therefore, probability of B is $1 - \pi$.

- The binomial distribution gives the probability that a specified outcome will occur in a given number of independent trials.

Example: Binomial

Suppose that the entire population of patients with HIV has been studied and the five year survival is 80%. If S represents a 5 year survival and D represents a death before 5 years, then

$$\pi = P(S) = 0.8$$
$$(1 - \pi) = P(D) = 0.2$$

Consider a group of **TWO** patients. What is the probability:

- — that exactly two patients survive for 5 years?
- — that one patient lives for 5 years?
- — that none live for 5 years?

Example: Binomial

From the multiplication rule for independent events:

- The probability that exactly two patients survive for 5 years:

$$P_1(S) \times P_2(S) = 0.8 \times 0.8 = 0.64$$

- The probability of one patient living for 5 years:

$$P_1(S) \times P_2(D) = 0.8 \times 0.2 = 0.16$$
$$P_1(D) \times P_2(S) = 0.2 \times 0.8 = 0.16$$
$$P(event\ 1\ or\ event\ 2) = 0.16 + 0.16 = 0.32$$

- The probability that neither survive:

$$P_1(D) \times P_2(D) = 0.2 \times 0.2 = 0.04$$

- The sum of all possible probabilities always sums to one.

$$Total\ probability = 0.64 + 0.32 + 0.04 = 1$$

Binomial Distribution

This same process can be applied for a group of patients of any size or for any number of trials. The probability of X outcomes in a group of size n if each outcome has the probability of π is

$$P(X) = \frac{n!}{X!\,(n-X)!}\pi^X(1-\pi)^{n-X}$$

where $n!$ means $1 \times 2 \times 3 \times 4 \times \cdots n$

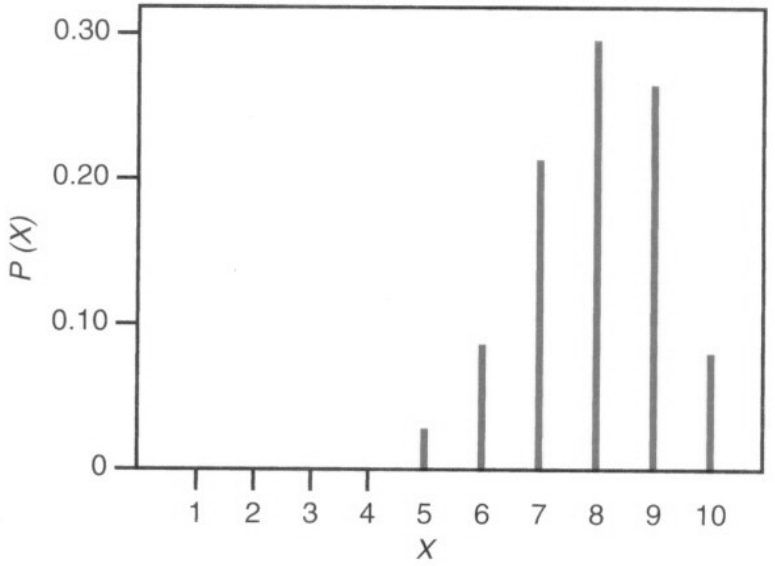
So for any new series of patients, we can use this formula to calculate the probability of X occurrences in n independent trials where the probability of success is π.

Example: Binomial

Suppose we had 10 HIV patients and $\pi = 0.8$ (P(S)). The probability that any number of them will survive can be calculated.

Number of Patients Surviving	$\frac{n!}{X!(n-X)!}$	π^X	$(1-\pi)^{n-X}$	$P(X)^a$
0	1	1	0.0000001	0
1	10	0.8	0.0000005	0
2	45	0.64	0.0000026	0.0001
3	120	0.512	0.0000128	0.0008
4	210	0.410	0.000064	0.0055
5	252	0.328	0.00032	0.0264
6	210	0.262	0.0016	0.0881
7	120	0.210	0.008	0.2013
8	45	0.168	0.04	0.3020
9	10	0.134	0.2	0.2684
10	1	0.107	1	0.1074

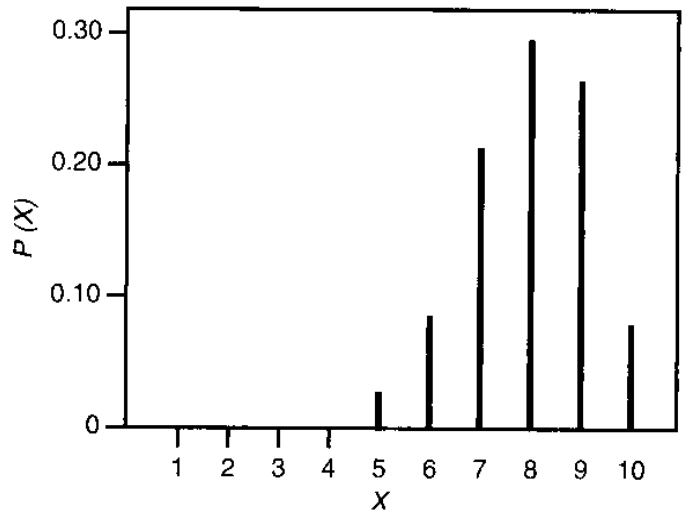
Rounded to four decimal places.



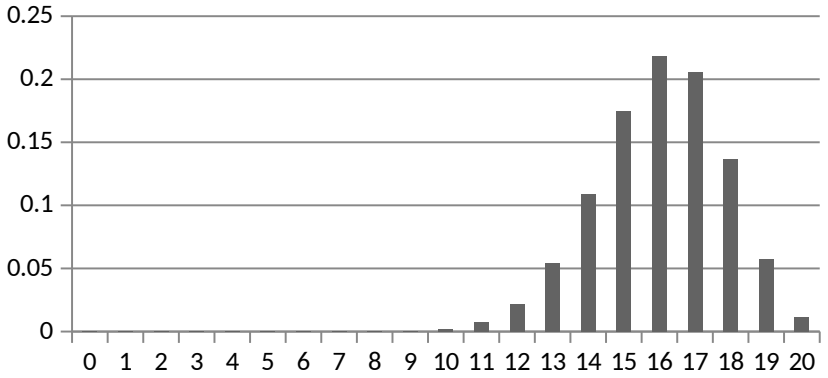
Binomial Approximation of Normal

- If the sample size is large, the normal distribution can be used.
- This will become useful when we do hypothesis testing.

n = 10 and π= 0.8
mean = nπ = 8



n = 20 and π = 0.8
mean = nπ = 16



Binomial Distribution

- Mean: $n\pi$
- Standard deviation: $\sqrt{n\pi(1-\pi)}$

- Thus, it is only necessary to know π and n to define the binomial distribution.

$$Binomial \sim (n, \pi)$$

- Often the interest is not the exact number of X successes but observing X or more successes.

- To calculate the probability of 8 or more people surviving 5 years:

$$\begin{aligned} P(X \geq 8) &= P(X = 8) + P(X = 9) + P(X = 10) \\ &= 0.3020 + 0.2684 + 0.1074 = 0.6778 \end{aligned}$$

In R

For sample size *n* and P(S) = p:

- $P(X \leq k) \rightarrow pbinom(k, n, p)$
- $P(X = k) \rightarrow dbinom(k, n, p)$
- $P(X \geq k) \rightarrow pbinom(k - 1, n, p, lower.tail = FALSE)$

Poisson Distribution

- When the outcome is the number of times an event occurs or to compute the probability of rare events. e.g. modelling the incidence of a disease or the number of cells growing in a medium.

- Like the binomial, probability of exactly X occurrences can be computed from a formula

$$P(X) = \frac{\lambda^X e^{-\lambda}}{X!}$$

where λ is the mean (e is a constant ≈ 2.71828).

$$Poisson \sim (\lambda)$$

Example: Poisson

- **390 individuals** living in a malaria-endemic region experienced a total of **1,256 malaria episodes** during a mean follow-up period of 11 years.

- Mean number of malaria episodes per individual (λ): $\frac{1256}{390} = 3.22$

The probability that an individual experiences zero malaria episodes during the follow-up period:

$$P(X = 0) = \frac{3.22^0 e^{-3.22}}{0!} = \frac{(1)(0.04)}{1} = 0.04$$

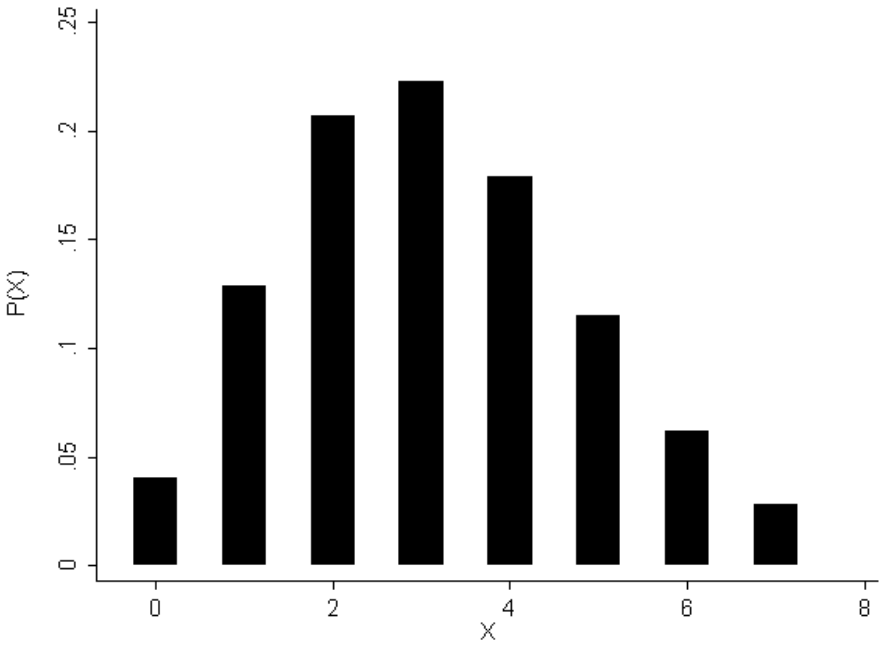
  - An individual has exactly one episode:

$$P(X = 1) = \frac{3.22^1 e^{-3.22}}{1!} = \frac{(3.22)(0.04)}{1} = 0.129$$

Poisson Distribution

- The calculations for the Poisson distribution when $\lambda = 3.22$ and $X = 0, 1, \ldots, 7$ are given in the table below:

No of hosp.	λ=3.22	$e^{-3.22}$	X!	P(X)
0	1	0.04	0	0.04
1	3.22	0.04	1	0.129
2	10.37	0.04	2	0.207
3	33.39	0.04	6	0.223
4	107.5	0.04	24	0.179
5	346.16	0.04	120	0.115
6	1114.64	0.04	720	0.062
7	3589.15	0.04	5040	0.028



In R

For mean m :

- $P(X \leq k) \rightarrow ppois(k, lambda = m)$
- $P(X = k) \rightarrow dpois(k, lambda = m)$
- $P(X \geq k) \rightarrow ppois(k - 1, lambda = m, lower.tail = FALSE)$

Normal (Gaussian) Distribution

- The characteristics of the normal distribution are:
  - continuous and can take on any value between $-\infty$ and $+\infty$, and not just integers
  - smooth; bell-shaped and symmetric around the mean μ
  - the standard deviation is denoted by σ

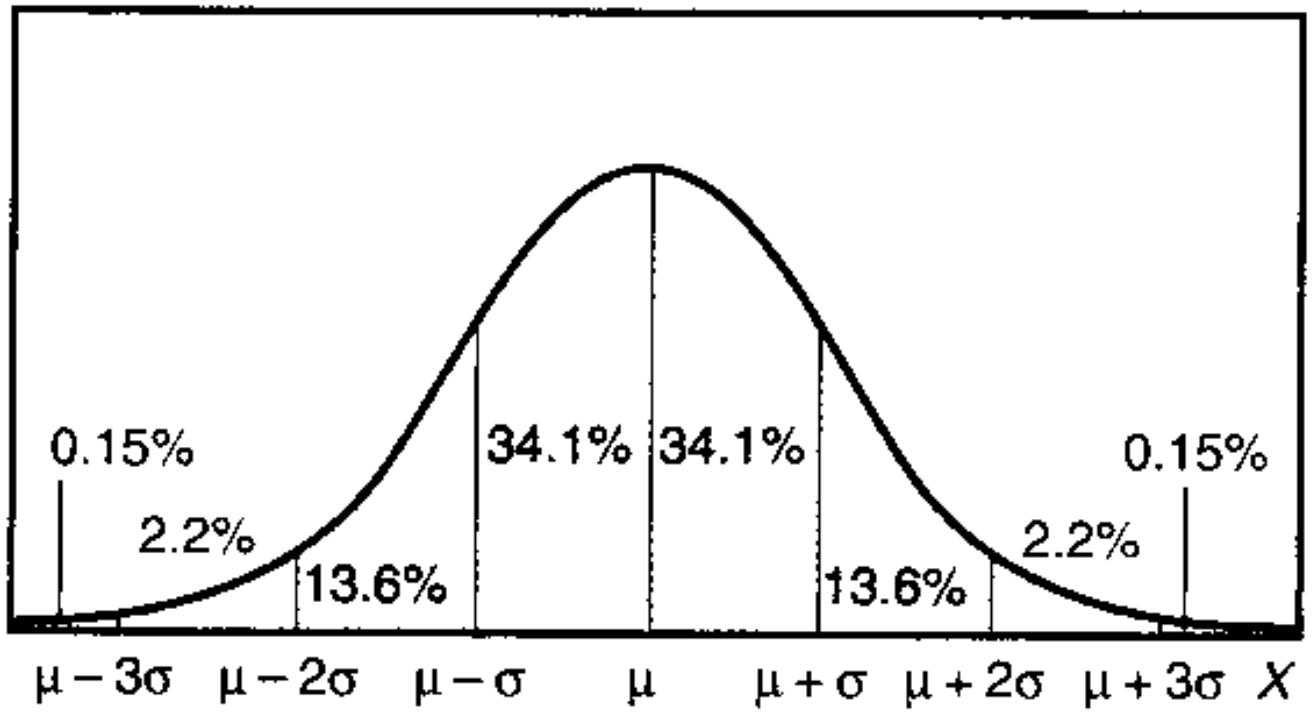
- The mean μ and standard deviation σ completely describe the normal distribution.

$$Normal \sim (\mu, \sigma^2)$$

- The area under the curve is equal to one because it is a probability distribution.

- Half the area lies to the left and half the area to the right of the mean.

Normal Curve



Since the area under the curve is symmetric and sums to one, the curve can be used to find the probability that an observation falls between *a* and *b*. The curve above shows some of the commonly used area.

Standard Normal Distribution

- To avoid using many different normal distributions for every μ and σ, the standard normal curve called z is used.

- The z-distribution has a mean of 0 and a standard deviation of 1.

$$Normal \sim (0, 1)$$

- A z-transformation can be used to convert any normal distribution into a standard normal curve.

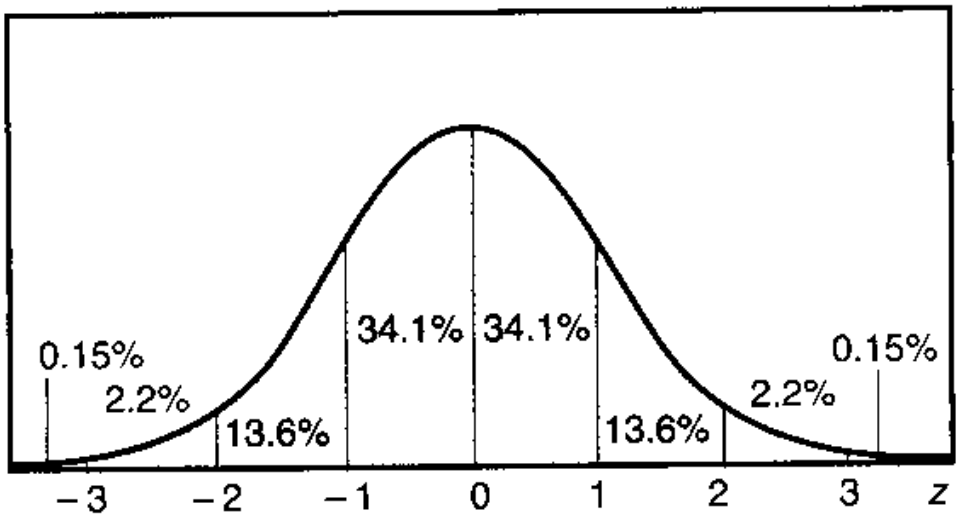
- Once transformed, statistical tables can be used to calculate the area under the curve.

- The z-transformation is

$$z = \frac{x - \mu}{\sigma}$$

- This is sometimes called a score, normal deviate, z-score.

Area under the Standard Normal Curve



What is the area

- Above 1?

  From the curve, the area is 13.6% + 2.2% + 0.15% = 15.95%

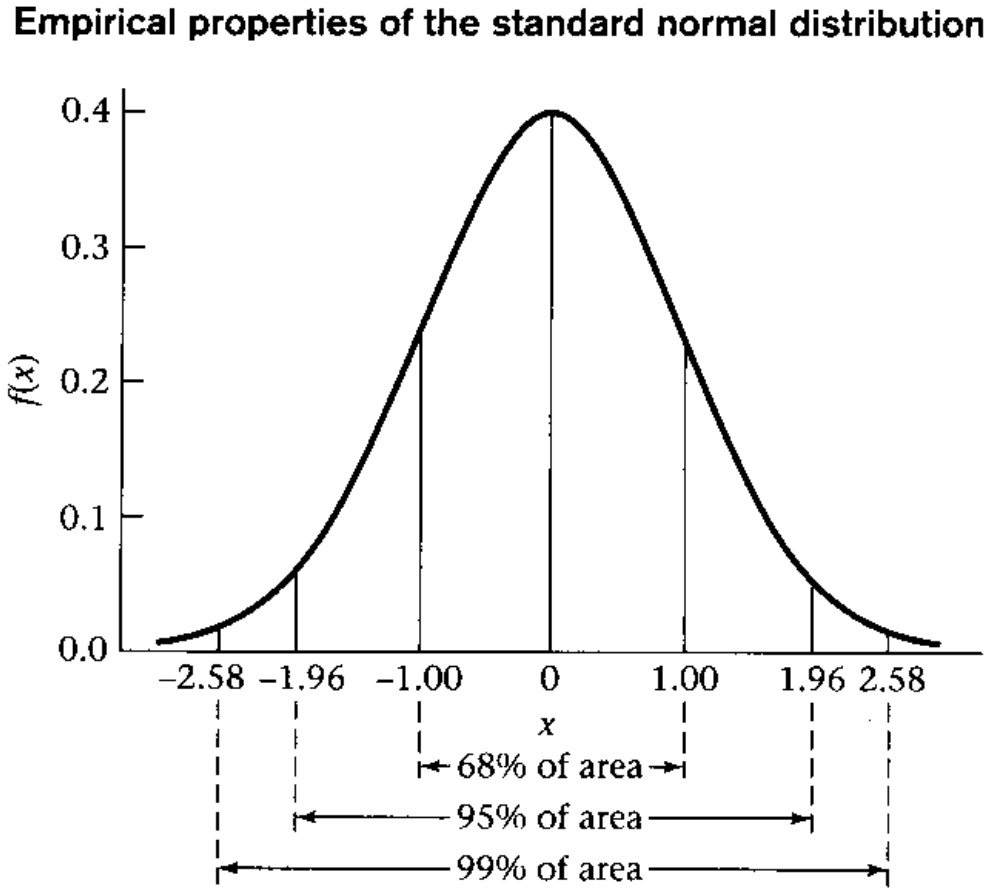
- Below -1?

  From the curve, the area is 0.15% + 2.2% + 13.6% = 15.95%

- Between -2 and 2?

  From the curve, 13.6% + 34.1% + 34.1% + 13.6% = 95.4%

Area under the Standard Normal Curve



Commonly used z-values and corresponding area

The z-table

Part of the table of area under a standard normal curve.

Areas under the standard normal curve.[a]

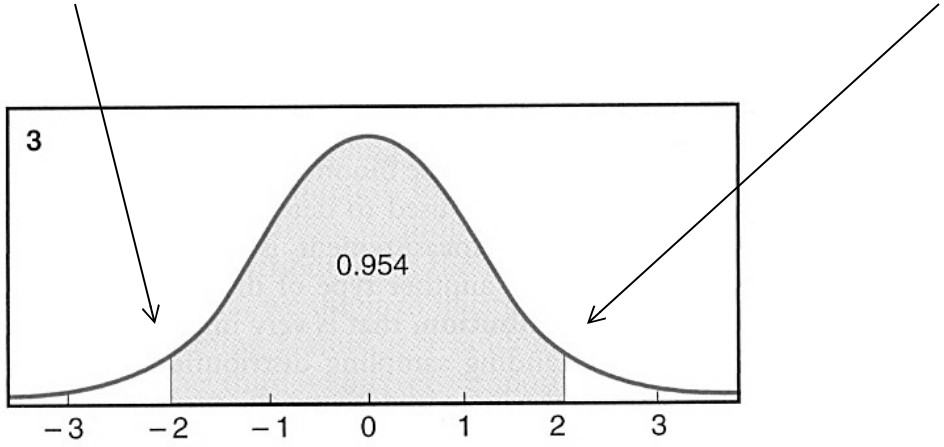
z	Area Between −z & +z	Area in Two Tails (< −z & > +z)	Area in One Tail (< −z or > +z)
0.00	0.000	1.000	0.500
0.05	0.040	0.960	0.480
0.10	0.080	0.920	0.460
0.15	0.119	0.881	0.440
0.20	0.159	0.841	0.421
0.25	0.197	0.803	0.401
0.30	0.236	0.764	0.382
0.35	0.274	0.726	0.363
0.40	0.311	0.689	0.345
0.45	0.347	0.653	0.326
0.50	0.383	0.617	0.309
0.55	0.418	0.582	0.291
0.60	0.451	0.549	0.274
0.65	0.484	0.516	0.258
0.70	0.516	0.484	0.242
0.75	0.547	0.453	0.227
0.80	0.576	0.424	0.212
0.85	0.605	0.395	0.198
0.90	0.632	0.368	0.184
0.95	0.658	0.342	0.171
1.00[a]	0.683	0.317	0.159
1.05	0.706	0.294	0.147
1.10	0.729	0.271	0.136
1.15	0.750	0.250	0.125
1.20	0.770	0.230	0.115
1.25	0.789	0.211	0.106
1.28[a]	0.800	0.200	0.100
1.30	0.806	0.194	0.097
1.35	0.823	0.177	0.089
1.40	0.838	0.162	0.081
1.45	0.853	0.147	0.074
1.50	0.866	0.134	0.067
1.55	0.879	0.121	0.061
1.60	0.890	0.110	0.055
1.645[a]	0.900	0.100	0.050
1.65	0.901	0.099	0.049
1.70	0.911	0.089	0.045
1.75	0.920	0.080	0.040
1.80	0.928	0.072	0.036
1.85	0.936	0.064	0.032
1.90	0.943	0.057	0.029
1.95	0.949	0.051	0.026
1.96[a]	0.950	0.050	0.025
2.00	0.954	0.046	0.023

Example: Standard Normal

How many people have a blood pressure between 100 and 140mm Hg?

- Calculate the area in the tail < 100mm Hg.

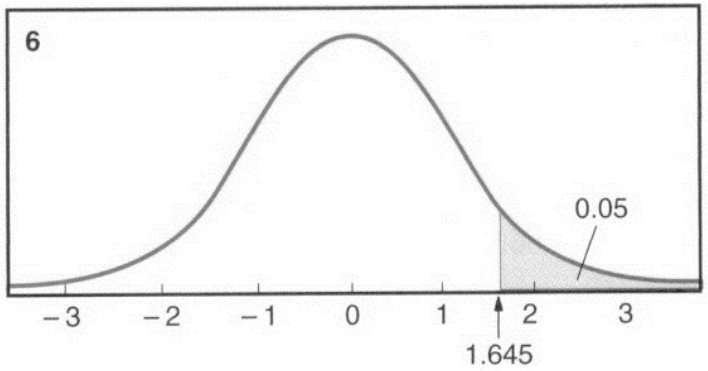
$$z_1 = \frac{100-120}{10} = -2 \qquad z_2 = \frac{140-120}{10} = 2$$



- The area between -2 and +2 is 0.954. So 95.4% have a systolic BP between 100 and 140mm Hg.

Example

What is the value of systolic BP that divides the area under curve into the lower 95% BP and upper 5%?



- The z-value from the table dividing the lower 0.95 from the upper 0.05, z = 1.645.

$$z = \frac{X-\mu}{\sigma} \text{ therefore } 1.645 = \frac{X-120}{10}$$

Solving for X, we get X = 136.45.

- A systolic BP of 136.45 is at the 95th percentile. So 95% of healthy people have a systolic BP of 136.45mm Hg or lower.

In R

dnorm() # Density: f(z)

pnorm() # Cumulative probability: P(Z ≤ z)

qnorm() # Quantiles (inverse normal)

rnorm() # Random normal values

Summary

- Binomial
  - used to model events with binary outcome to determine the probability of outcomes of interest
- Poisson
  - used to model the number of times an event occurs or the probability of rare events
- Normal
  - used to determine the probability of characteristics measured on a continuous scale
  - when the distribution is approximately bell-shaped
  - shows how representative or extreme an observation is

Summary

Features of the normal distribution:

1. The mean ± 1 standard deviation contains approximately 66.7% of the area under the normal curve.

2. The mean ± 2 standard deviations contains approximately 95% of the area under the normal curve.

3. The mean ± 3 standard deviations contains approximately 99.7% of the area under the normal curve.